**NATALIE MCCLAIN, BA**
natalie.mcclain@icahn.mssm.edu • (502) 303-4101

## EDUCATION

**Barnard College of Columbia University** *Sep 2018 – May 2022*
B.A. in Neuroscience & Behavior (Cellular) | Minor: Mathematics | *Cum Laude*

**Icahn School of Medicine, Neuroscience PhD Program** *Aug 2024 – Present*
Mentor: Rita Goldstein, PhD | NRSA TL1 recipient

## RESEARCH EXPERIENCE

**Neuropsychoimaging of Addiction and Related Conditions Research Program, Icahn School of Medicine at Mount Sinai** *Aug 2020 – Present*

Neuroscience PhD Candidate *(Aug 2024 – Present)*

- Exploring brain responses to a drug-themed movie as a naturalistic marker of drug-biased processing, as a function of treatment status, hormonal fluctuations, and drug use.
- Studying predictive validity of behavioral markers in heroin use disorders, as compared with traditional self-reported severity measures.
- Investigated functional connectivity during extinction of drug memories under methylphenidate in individuals with cocaine use disorder.

Clinical Research Coordinator *(Jul 2022 – Aug 2024)*

- Ran neuroimaging protocols (MRI, PET/MR, EEG), conducted all participant-facing study procedures (recruitment, screening, behavioral tasks, psychometric surveys, interventions), and coordinated clinical procedures (neurology exams, carotid ultrasounds, cardiovascular screening, reviewing blood results).
- Investigated behavioral measures as predictors of prospective treatment adherence as compared to self-reported severity metrics in individuals with heroin use disorder with postdoctoral fellow Ahmet Ceceli, Ph.D.
- Coordinated and assisted with grant submissions, annual reporting to the NIH, and IRB correspondence.

Research Assistant *(Aug 2020 – Jul 2022)*

- Investigated spoken recall of a drug movie as a naturalistic measure of impaired salience attribution in individuals with heroin use disorder with postdoctoral fellow Ahmet Ceceli, Ph.D.
- Conducted the participant-facing procedures as listed above.

**Summer Research Institute, Barnard College** *May 2021 – September 2021*

- Received summer funding and supportive programming to investigate spoken recall of a drug movie as an indicator of attentional bias in a naturalistic context in individuals with heroin use disorder.

**Losonczy Lab, Zuckerman Institute of Columbia University** *May 2019 – Jan 2021*

Research Assistant

- Investigated place cell tuning in the mouse hippocampus and its role in spatial memory by examining dendritic membrane potential dynamics using voltage imaging and local field potential recording.
- Performed neuron segmentations and morphology classification, *in vivo* electroporation of CA1 pyramidal cells under two-photon microscopy guidance, and plasmid preparation.

## AWARDS & RECOGNITIONS

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| Neuroscience Scholars Program Associate, Society for Neuroscience ($500) | 2026 |
| Summer Program in Neuroscience, Excellence and Success ($1400, *declined*) | 2026 |
| New Investigator Award, American Society of Clinical Psychopharmacology ($1500) | 2026 |
| Graduate Student Travel Award, Icahn School of Medicine ($800) | 2025, 2026 |
| NRSA TL1 Pre-Doctoral Career Development Training Award | 2025-2026 |
| Mount Sinai Trainee Innovation Award ($600) | 2025 |
| Prize for Distinguished Achievement in Neuroscience, Barnard College | *2022* |
| Summer Research Institute Grant, Barnard College ($5,000) | *2021* |
| Summer Research Stipend, Losonczy Lab, Zuckerman Institute ($4,000) | *2020* |
| Dean's List, Barnard College | *2018-2022* |

## PUBLICATIONS

Peer-Reviewed Journal Articles

1. Ceceli AO, King SG, **McClain N,** Alia-Klein N, Goldstein RZ. (2023) The Neural Signature of Impaired Inhibitory Control in Individuals with Heroin Use Disorder. *Journal of Neuroscience*, 43(1), 173-182.
2. Ceceli AO, Huang Y, Kronberg G, Malaker P, Miller P, King SG, Gaudreault PO, **McClain N,** Gabay L, Vasa D, Newcorn JH, Ekin D, Alia-Klein N, Goldstein RZ. (2023) Common and distinct fronto-striatal volumetric changes in heroin and cocaine use disorders. *Brain*, 146(4), 1662–1671. (Recognized in *Faculty Opinion* as exceptional new finding; Recognized in *Brain* editorial)
3. Ceceli AO, Huang Y, Gaudreault PO, **McClain NE,** King SG, Kronberg G, Brackett A, Hoberman GN, Gray JH, Garland EL, Alia-Klein N, Goldstein RZ. (2024). Recovery of inhibitory control prefrontal cortex function in inpatients with heroin use disorder: A 15-week longitudinal fMRI study. *Nature Mental Health*, 2(6), 694-702.
4. Liao Z, Gonzalez KC, Li DM, Yan CM, **McClain NE**, Zhang G, Evans SW, Chavarha M, Simko J, Makinson CD, Lin MZ, Losonczy A, Negrean A. (2024) Functional architecture of intracellular oscillations in hippocampal dendrites. *Nature Communications,* 15(1).
5. Kronberg G*, Ceceli AO*, Huang Y, Gaudreault PO, King SG, **McClain N,** Alia-Klein N, Goldstein RZ. (2024) Naturalistic drug cue reactivity in heroin use disorder: orbitofrontal synchronization as a marker of craving and recovery. *Brain,* awae369.
6. Huang Y, Butelman E, Ceceli AO, Kronberg G, King SG, **McClain NE**, Wong Y, Boros M, Drury K, Sinha R, Alia-Klein N, Goldstein RZ. (2025) Sex and hormonal effects on drug cue-reactivity and its regulation in human addiction. *Biological Psychiatry*.
7. Ceceli AO, King S, Drury K, **McClain N**, Gray J, Dassanayake PS, Newcorn J, Schiller D, Alia-Klein N, Goldstein R. (2025) The neural signature of methylphenidate- and reconsolidation-enhanced memory disruption in human drug addiction. *PNAS*.
8. Ceceli AO, Huang Y, Kronberg G, **McClain N**, King S, Butelman E, Alia-Klein N, & Goldstein RV. (2025) The impaired Response Inhibition and Salience Attribution (iRISA) model of drug addiction: recent neuroimaging evidence and future directions. *Annual Reviews Psychology*.
9. **McClain N*,** Ceceli AO*, Kronberg G, Alia-Klein N, Goldstein RZ. (2025) Moving beyond self-report in characterizing drug addiction: Using behavioral measures to inform treatment adherence. *Biological Psychiatry: Global Open Science*. (*equal contribution)
10. Huang Y, Ceceli AO, Kronberg G, Drury KR, King SG, **McClain NE**, Wong YY, Boros M, Butelman ER, Gaudreault PO, Parvaz MA, Alia-Klein N, Garland EL, Goldstein RZ. (2026) Mindfulness-Oriented Recovery Enhancement rebalances prefrontal responses to drug and natural reward cues in opioid use disorder. *JAMA Psychiatry* (In press)
11. Moeller SJ, **McClain N**, Abeykoon S, Alia-Klein N, Goldstein RZ. (2026) Metacognitive impairment in cocaine use disorder: Confirmatory evidence for the effects of recent drug use. *Translational Psychiatry.* (In press)

Preprints

1. King S, Kronberg G, **McClain N,** Zhang Z, Ceceli A, Huang Y, Alia-Klein N, Goldstein R. (Under revision in *Journal of Neuroscience*). Verbal recall of a drug-themed movie captures treatment effects and neural encoding in heroin use disorder.
2. Kronberg G*, **McClain N***, Hennings AC, Hasson U, Alia-Klein N, Norman K, Goldstein R. (In preparation). Peer-guided movie neurofeedback to facilitate recovery in opioid use disorder. (*equal contribution)
3. Ceceli A, Kronberg G, **McClain N,** Daw N, Alia-Klein N, Goldstein RZ. (In preparation) Motivational control tracks vulnerability during recovery in human addiction.
4. **McClain N*,** Huang Y*, Kronberg G, Alia-Klein N, Goldstein R. (In preparation) Menstrual phase and hormonal effects on drug-biased synchronization in women with cocaine addiction.

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## PRESENTATIONS

Oral Presentations

1. **McClain N*,** Kronberg G*, Alia-Klein N, Goldstein R. (2026) Brain-to-brain neurofeedback to facilitate recovery in opioid use disorder. *American Society of Clinical Psychopharmacology*; Miami, FL.
2. **McClain N**, Ceceli A, Kronberg G, Drury K, Alia-Klein N, Goldstein RZ. (2026) Altered brain state dynamics during a drug-themed movie in opioid use disorder. *Gordon Research Seminar: Neurobiology of Addiction*; Barcelona, Spain. (Also a poster presentation)
3. Huang Y*, **McClain N***, Kronberg G, Butelman ER, Ceceli AO, King SG, Wong YY, Boros M, Drury R, Sinha R, Alia-Klein N, Goldstein RZ. (2026) Sex Differences and Hormonal Effects in Drug Cue-reactivity and its Regulation in Human Drug Addiction. *American College of Neuropsychopharmacology*; Nassau, Bahamas.

4. King S, Kronberg G, **McClain N**, Zhang N, Ceceli A, Gray J, Alia-Klein N, Goldstein RZ. (2023) Verbal movie recall reveals heightened self-reference in individuals with heroin use disorder. Society for Neuroscience 2023; Washington, DC. (Also a poster presentation)
5. **McClain N** & Ceceli AO**.** (2022) Impaired Salience Attribution in Drug Addiction. Barnard College Neuroscience & Behavior Senior Thesis Conference; New York, NY, 2022.

Published Abstracts

1. Kronberg, G., Ceceli, A. O., **McClain, N**., Vasa, D., Malaker, P., Alia-Klein, N., & Goldstein, R. Z. (2021). Inter-Subject Correlations of fMRI Responses to Cinematic Heroin Use in Individuals with Heroin Use Disorder. *Neuropsychopharmacology*, 47, pp. 425-426. (Presented at American College of Neuropsychopharmacology 2021; San Juan, PR and Society for Neuroscience 2021; Virtual)
2. Ceceli AO, King S, Kronberg G, **McClain N,** Vasa D, Alia-Klein N, Goldstein RZ. Speech and viewing choice as naturalistic markers of drug-biased salience attribution in individuals with heroin use disorder. *Neuropsychopharmacology*, 46(1), pp. 410. (Presented at American College of Neuropsychopharmacology 2022; Phoenix, AZ).
3. Kronberg G, Ceceli AO, Huang Y, Gaudreault PO, **McClain N,** Vasa D, Malaker P, Ekin D, Alia-Klein N, Goldstein RZ. (2022). P37. Effects of Abstinence During Treatment on Shared Nucleus Accumbens Reactivity and Craving in Response to a Drug-Related Movie in Heroin Addiction. *Biological Psychiatry*, 91(9), S102-S103. (Presented at Society of Biological Psychiatry 2022*;* New Orleans, LA)

Poster Presentations

1. **McClain N** & Ceceli AO. Investigating impaired salience attribution in drug addiction via naturalistic stimuli: Speech encoding for drug- and non-drug-related movie scenes in individuals with heroin addiction. Barnard College Lida Orzeck ’68 Poster Session 2021; New York, NY.
2. Ceceli AO, King S, **McClain N,** Gaudreault PO, Alia-Klein N, Goldstein RZ. The neural signature of impaired inhibitory control and its recovery in heroin addiction. Gordon Research Conference 2022; Orlando, FL.
3. King S, Kronberg G, **McClain N**, Ceceli A, Gray J, Alia-Klein N, Goldstein RZ. Verbal movie recall reveals heightened self-reference and contextual variability in heroin-addicted individuals. Columbia University Curiosity, Creativity, and Complexity Conference 2023; New York, NY.
4. Huang Y, Ceceli AO, Kronberg G, King S, Brackett A, Hoberman GH, **McClain NE**, Alia-Klein NE, Goldstein RZ. Sex differences in cue-reactivity and reappraisal in heroin and cocaine use disorder — an fMRI study. Society for Neuroscience 2023; Washington, DC.
5. Huang Y, Butelman ER, **McClain N,** Hao Q, Dhamoon M, Robson P, Kaufman A, Goldstein RZ, Fayad Z, Alia-Klein N. Multi-modal imaging of carotid atherosclerotic risk in substance use disorders. Biomedical Engineering and Imaging Institute Symposium 2024; New York, NY.
6. **McClain, N*,** Ceceli, AO*, Kronberg, G, Alia-Klein, N, Goldstein, RZ. (*equal contribution) Beyond self-report in characterizing drug addiction: Using behavior to inform treatment adherence. Friedman Brain Institute Retreat 2024; New York, NY.
7. Ceceli AO, King S, Drury K, **McClain N**, Gray J, Newcorn J, Schiller D, Alia-Klein N, & Goldstein RZ. Disrupting drug memories to diminish drug cue-reactivity using reconsolidation and methylphenidate in addiction: A double-blind randomized clinical trial in individuals with cocaine use disorder. American College of Neuropsychopharmacology 2024; Phoenix, AZ.
8. **McClain, N,** Drury, K, Kronberg, G, Alia-Klein, N, Goldstein, RZ. Reduced Synchronization of Neural Responses to a Drug-Related Movie in Opioid Use Disorder. Friedman Brain Institute Retreat 2025; New York, NY.
9. **McClain, N,** Kronberg, G, Drury, K, Alia-Klein, N, Goldstein, RZ. Menstrual Phase Effects on Brain State Dynamics during a Drug-Themed Movie in Cocaine Addiction. American College of Neuropsychopharmacology 2026; Nassau, Bahamas.
10. **McClain N**, Ceceli A, Kronberg G, Drury K, Alia-Klein N, Goldstein RZ. (2026) Altered brain state dynamics during a drug-themed movie in opioid use disorder. Friedman Brain Institute Retreat 2026; New York, New York and Gordon Research Seminar and Conference: Neurobiology of Addiction; Barcelona, Spain

## TRAINING PROGRAMS

**Neurohackademy: Summer Program in Neuroimaging and Data Science** *Aug 2025*
University of Washington eScience Institute

**National Science Foundation Regional Innovation Corps** *Feb 2026*
Icahn School of Medicine at Mount Sinai

**New York Computational Psychiatry Workshop** *Oct 2024*
Mount Sinai Center for Computational Psychiatry

## TEACHING

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| Teaching Assistant | Fundamentals of BMS Core IV: Neuroscience (Spring 2026) |
| Teaching Assistant | Introduction to Scientific Programming in Python (Fall 2026) |
| Course Assistant | Neuroscience Department, Barnard College (Fall 2026) |

## MENTORSHIP EXPERIENCE

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| Shraeya Santhapuram | MA student in Psychology at NYU |
| Christy Garlock | MA student in Psychology at NYU |
| Maggie Boros | Research Coordinator at NARC Lab |
| Rachel Drury | Research Coordinator at NARC Lab |
| Chloe Wong | Research Coordinator at NARC Lab |

## VOLUNTEERING

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| OnPoint NYC | Fall 2025 – present |
| Crown Heights Community Kitchen | Fall 2025 – present |